

Research Notes on DenseNet

DenseNet

>> Section 1

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3×3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer:

>> Section 2

Pre-activation block The pre-activation residual block applies activation functions before applying the residual function F . Formally, the computation of a pre-activation residual block can be written as:

>> Section 3

Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3×3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

>> Section 4

$$\begin{aligned} \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} &= \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} \frac{\partial x}{\partial L} \frac{\partial x}{\partial L} \frac{\partial x}{\partial L} = \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} (1 + \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x} \frac{\partial x}{\partial L} - 1 F(x_i)) \\ &= \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} + \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} \frac{\partial x}{\partial L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x} \frac{\partial x}{\partial L} - 1 F(x_i) \\ &\frac{\partial E}{\partial x} \frac{\partial x}{\partial L} = \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} + \frac{\partial E}{\partial x} \frac{\partial x}{\partial L} \frac{\partial x}{\partial L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x} \frac{\partial x}{\partial L} - 1 F(x_i) \end{aligned}$$

>> Section 5

During the late 1980s, "skip-layer" connections were sometimes used in neural networks. Examples include: Lang and Witbrock (1988) trained a fully connected feedforward network where each layer skip-connects to all subsequent layers, like the later DenseNet (2016). In this work, the residual connection was the form $x \mapsto F(x) + P(x)$, where P is a randomly-initialized projection connection. They termed it a "short-cut connection". An early neural language model used residual connections and named them "direct connections".

>> Section 6

where ϕ can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings

between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

>> Section 7

Forward propagation If the output of the ℓ -th residual block is the input to the $(\ell + 1)$ -th residual block (assuming no activation function between blocks), then the $(\ell + 1)$ -th input is:

>> Section 8

Backward propagation The residual learning formulation provides the added benefit of mitigating the vanishing gradient problem to some extent. However, it is crucial to acknowledge that the vanishing gradient issue is not the root cause of the degradation problem, which is tackled through the use of normalization. To observe the effect of residual blocks on backpropagation, consider the partial derivative of a loss function E with respect to some residual block input x_{ℓ} . Using the equation above from forward propagation for a later residual block $L > \ell$:

>> Section 9

Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c_t that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.

>> Section 10

This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x} \frac{\partial x}{\partial L}$, always has a later term $\frac{\partial E}{\partial x} \frac{\partial x}{\partial L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x} \frac{\partial x}{\partial L}$

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$\frac{\partial E}{\partial x_L}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L}$.

>> Section 11

A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of $x \mapsto f(x) + x$, where f is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

>> Section 12

Residual connection In a multilayer neural network model, consider a (non-residual) subnetwork with a certain number of stacked layers (e.g., 2 or 3). Let $H(x; \alpha)$ denote the subnetwork. Suppose H^* is the desired optimal output of this subnetwork. Residual learning simply adds x directly to the output, such that the optimal learned output now becomes $H^* - x$, which is interpreted as a "residual" with respect to x . The operation of "adding x " is implemented via a "skip connection" that performs an identity mapping to connect the input of the subnetwork with its output. This connection is referred to as a "residual connection" in later work. Let $F(x; \alpha) = H(x; \alpha) + x$. The function F is often represented by matrix multiplication interlaced with activation functions and normalization operations (e.g., batch normalization or layer normalization). As a whole, one of these subnetworks is referred to as a "residual block". A deep residual network is constructed by simply stacking these blocks. Long short-term memory (LSTM) has a memory mechanism that serves as a residual connection. In an LSTM without a forget gate, an input x_t is processed by a function F and added to a memory cell c_t , resulting in $c_{t+1} = c_t + F(x_t)$. An LSTM with a

forget gate essentially functions as a highway network. To stabilize the variance of the layers' inputs, it is recommended to replace the residual connections $x + f(x)$ with $x/L + f(x)$, where L is the total number of residual layers.